The Journal of FINANCE
The Journal of THE AMERICAN FINANCE ASSOCIATION

THE JOURNAL OF FINANCE • VOL. LXXX, NO. 5 • OCTOBER 2025

Baby Booms and Asset Booms: Demographic Change and the Housing Market

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ABSTRACT

Based on centuries of data, we demonstrate that demographics have been a major, predictable driver of house prices. High birth rates 25 to 29 (60 to 64) years ago predict declining (rising) rent-price ratios today. This pattern arises from age-concentrated entry into and exit from homeownership affecting house prices, rather than changes in housing consumption that could also impact rents. We provide evidence for possible mechanisms: slow responses of other market participants to shifts in homeownership demand, and geographic segmentation between rental and owner-occupied markets. Evidence for age-dependent demand effects on yields of bonds and stocks is significantly weaker.

IN MOST DEVELOPED COUNTRIES, INDIVIDUALS' housing trajectories follow a similar pattern. Initially, children live with their parents. By their late teens or early 20s, many move to rental properties. By the time individuals start settling down, often in their mid- to late 20s and onward, they start looking for owner-occupied housing. Only at old ages do some individuals transition back to rental housing. This housing life cycle could influence the housing market and price disparities between owned and rented properties. Notably, a large U.S. cohort around age 30, a typical age for property acquisition, has recently entered the market for owner-occupied housing, and some suggest this influx

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DOI: 10.1111/jofi.13480

may have contributed to the post-COVID-19 housing boom.[1] At the same time, the aging postwar baby boomers will eventually shift to elderly housing or pass away.

The purpose of this paper is to examine the impact of such demographic changes on the housing market. We show that age-dependent entry into and exit out of homeownership is a key and predictable driver of home prices, explaining about a fifth of the variation in rents relative to house prices between the 17^{th} and 19^{th} centuries. Birth rate shocks predict large house price increases at the time a cohort enters its prime home-buying years and decreases when a cohort exits homeownership at old age or after death. We find no or limited evidence for such effects on rent prices or the yields of other assets. This effect is specific to housing investment because shifts in demand for homeownership are much more age-concentrated than shifts in demand for housing consumption or other financial assets. Differences in cohort size can cause large imbalances between homes available for sale and the number of searching homebuyers. This might have large price effects if other market participants do not enter the market sufficiently quickly to change the relative supply of owner-occupied and rental property.

After introducing the data and historical context, this paper starts with a set of three stylized facts on housing investment and demographics that motivate our main analysis and research question. First, housing purchases and sales are highly age-concentrated, with net housing purchases peaking for individuals in their late 20s and 30s, while net housing sales concentrate at older ages. Linking microdata on housing transactions and the age of buyers and sellers, this pattern holds both today and centuries ago. There are some differences in the exact distribution of buyers' and sellers' ages across markets, likely related to differences in the amount of equity required to buy a house, difference in ownership structure, and differences in cohort sizes.

Second, housing consumption demand is more smooth over adulthood, rising most in adolescence and early adulthood when children move out of their parental homes. Many individuals rent in early adulthood before buying, and homebuyers sometimes transition back to renting at older ages. Survey data also indicate that the majority of individuals prefer to own their house when they are between ages 25 and 65, whereas they prefer to rent before and after. In short, the strong peak in realized housing purchases in individuals' late 20s and 30s and the peak in sales for seniors is likely related to changing tenure preferences.

Third, we document substantial time variation in mortality and fertility. While the 20^{th} century was characterized by declining fertility and a single large baby boom, historically baby booms and busts happened more frequently. Structural shifts in mortality and fertility in the 19^{th} and 20^{th} centuries also imply that the impact of a given demographic shock has changed. For example,

[1] See, for example, "Millennials are Supercharging the Housing Market," *Wall Street Journal*, December 2021, and "Millennial Demand Helps Stoke the Housing Boom," *The Economist*, March 2022.

increasing life expectancy implies that many more people are likely to own property and demand housing at older ages, and that few children die before they reach adulthood.

Although age patterns in demand for homeownership and housing consumption are clear, it is much more difficult to link changes in cohort size to changes in housing market outcomes. Empirical identification has been challenging since the onset of this literature (e.g., Mankiw and Weil (1989), Poterba (2001)), and thus most papers have relied on theoretical models. On the one hand, demographic changes tend to be very gradual and large baby booms rare, implying that one needs long-horizon data to study its impact. On the other hand, virtually all economies have experienced sizable changes in fertility and mortality over time, as well as substantial economic growth. This makes it very difficult to study the effect of demographic shocks, since cohorts vary not only by size but also by life expectancy and the economic conditions they have faced.

The historical period we examine, from the 16$^{\text{th}}$ century to the 19$^{\text{th}}$ century, allows us to overcome these challenges. The period occurs before the demographic transition when fluctuations in fertility, mortality, and economic growth are less predictable. Moreover, the period is long enough to help alleviate concerns about serial correlation. To examine whether past birth rates are able to explain current changes in the housing market, we combine existing and newly constructed time series of demographic rates and asset returns with historical microdata on housing and investment decisions. Our study is based mainly on data from Amsterdam, but also from Paris (rents only). For these two cities, enough data survived to construct long time series.

In our main analysis, we examine the extent to which different lags of past birth rates can explain current changes in rents relative to house prices (rent-price ratios), focusing on lags corresponding to birth rate cohorts from adolescence until old ages. This empirical approach has important advantages for identification. First, by looking over very long horizons in an environment with no structural changes in demographic rates and limited economic growth, we are able to isolate the predictive impact of demographic rate shocks on housing costs. Due to structural shifts in demographic rates and the economy during the 19$^{\text{th}}$ and 20$^{\text{th}}$ centuries, as well as significant government interference in the housing market around the two World Wars, this is difficult to accomplish with shorter-term data and when extending the data to more recent times. Second, by using lagged birth rates rather than current demographic structure, we are able to circumvent the endogeneity of the latter, as migration will correlate both with economic opportunity and the share of younger households, particularly in cities. This design choice also allows to test for predictability. Finally, to rule out the possibility that our effects are driven by unobserved correlation between past birth rates and current economic outcomes, we focus on the relative changes of house prices and rents. Even if past birth rates correlate with current economic outcomes, it is unclear why this would affect house prices differently than rents beyond their direct effects on the housing market.

As our main result, we find that a one-percentage-point increase in the total five-year birth rate 25 to 29 years ago increases five-year growth in house prices relative to rents by about 4%, while we find the exact opposite for birth rates 60 to 64 years ago, when the effect is negative. Supporting our identification strategy, we find that these effects do not change when we gradually incorporate a wide set of current and lagged economic and demographic control variables. These effects are economically important: these two birth rate lags alone explain almost a fifth of the total variation in rent-price ratios over a 250-year horizon. When we look at changes in house prices and rents individually, we find that the effects concentrate in house prices and that demographic changes explain little to none of the variation in rent prices. These results suggest that age-dependent entry into homeownership is a key driver of changes in house prices relative to rents. In line with this view, the timing of our effects coincides with the ages at which people in the data typically start looking for homeownership or exit from it.

To better understand this particular mechanism, and to rule out alternative mechanisms, we conduct several additional analyses. Our intuition is that large price effects can be rationalized if other market participants respond only weakly to changes in the demand for homeownership by converting rental units to owner-occupied units and vice versa. While we cannot infer this directly from the data, we perform two exercises to shed light on whether this is the case.

We first look at the effects on sales rates, estimating the effect of lagged birth rates on the probability of a sale. In a given year, only a small fraction of the housing stock trades hands, and if other market participants do not put more properties on the market when demand for homeownership increases, it might result in significant price effects. We find that the arrival of a large birth cohort at home-buying age and their departure at old age increases the probability of a sale, but this effect is slow to materialize, occurring several years after we observe price effects.

We then analyze whether segmentation between markets for owner-occupied property and rental property could contribute to the price effects we find. Based on a rental census available for most of Amsterdam, we compute homeownership rates at the street level and estimate separate house price indices for both low and high homeownership areas. We find that relative changes in house prices between low and high homeownership areas can be predicted by past birth rates, in line with demographic demand profiles. For birth rate lags corresponding to cohorts in their late teens or early 20s, the typical age at which individuals start renting, we find higher house price growth in areas with many rental properties. Conversely, we find higher house price growth in areas with high homeownership for birth rate lags corresponding to adults in their early 30s. For birth rate lags corresponding to older individuals, we find fewer price declines in low relative to high homeownership areas, in line with some older individuals returning to rental property.

The two analyses above provide suggestive evidence that frictions in the housing market can contribute to age-dependent entry into and out of homeownership having large price effects. We next examine whether alternative explanations could play a role too. First, we study the extent to which large effects on house prices can persist due to constraints on new construction. Comparing periods with and without significant supply constraints, we show that our main price results are not driven by periods with supply constraints. Second, we examine whether the effects that we find on rent-price ratios also appear for other assets. If we find similar effects in other assets, that would suggest our effects are explained by general life-cycle demand for assets rather than entry and exit into homeownership. Looking at very long-term series of bond yields and dividend yields, we find no effects of birth rates 25 to 29 years ago on the prices of other assets, and weak and less consistent evidence for a positive but much smaller effect on bond and dividend yields for birth rates 60 to 64 years ago. A potential explanation for the latter effect is that asset sales of heirs after death might also put some pressure on the yields of other assets.

In conclusion, the large price effects that we document appear to be explained predominantly by the fact that the market for owner-occupied housing is generally slow to respond to large demand shocks. While other market participants ultimately respond to such changes, this tends to take place only several years later, after prices have adjusted. Because age is such a strong predictor of homeownership demand, this results in demographics being a key driver of changes in house prices. This appears not to be a mere artifact of the historical context we study. To test whether these results have some external validity, we perform a similar but less precisely identified analysis for modern Organization for Economic Co-operation and Development (OECD) countries in Section VI of the Internet Appendix.[2] The correlational findings of this analysis are very similar to what we document in historical Amsterdam, implying that these mechanisms likely continue to play an important role in the housing market today.

Related Literature and Contribution

Our findings contribute to existing work in both finance and urban economics. First, our paper relates to the literature investigating the link between demographic changes and asset prices. A large stream of theoretical papers uses overlapping-generations models to calibrate the impact of demographic shocks on interest rates or asset prices (e.g., Abel, 2001, 2003; Geanakoplos, Magill, and Quinzii, 2004; Carvalho, Ferrero, and Nechio, 2016; Eggertsson, Mehrotra, and Robbins, 2019; Kopecky and Taylor, 2020; Leombroni et al., 2020; Auclert et al., 2021; Gagnon, Johannsen, and López-Salido, 2021). In line with our empirical approach, several of these papers assume that demographic shifts originate from shocks to birth rates. Although there is no strong consensus from this literature, results generally point to large young cohorts having positive impacts on risky asset prices, while large old cohorts depress rates and increase risk premia. On the empirical side, Poterba (2001, 2004) generally finds limited support that demographic shifts affect

[2] The Internet Appendix may be found in the online version of this article.

aggregate asset prices, while Favero, Gozluklu, and Tamoni (2011) find that the middle-aged to young ratio does predict dividend-price ratios.

At the individual stock level, DellaVigna and Pollet (2007) find that demographics predict stock returns in industries with age-dependent demand. Such demographic shifts also affect capital budgeting decisions and cash holdings (DellaVigna and Pollet, 2013; Cunha and Pollet, 2020). These papers exploit the fact that differences in cohort sizes lead to changes in demand, which investors do not appear to anticipate despite these changes being highly predictable. Our paper builds on a similar idea, namely, that past shocks to birth rates lead to predictable changes in demand for homeownership today, and that these result in price effects as investors do not appear to anticipate them in time.

We differ from most of the finance literature by explicitly focusing on housing assets, which is households' main investment asset.[3] Housing is lumpy and illiquid, and net buying or selling correlates more strongly with age than investments in other assets. Accordingly, the large impact of demographics on house prices and rent-price ratios that we document is likely specific to the housing market. Consistent with this view, we do not find strong empirical evidence for such effects in other assets.

Starting with Mankiw and Weil (1989), various papers look at the relation between demographic changes and housing costs (e.g., Engelhardt and Poterba, 1991; Green and Hendershott, 1996; Takáts, 2012; Green and Lee, 2016; Hiller and Lerbs, 2016; Gong and Yao, 2022). These papers establish how observed housing consumption demand varies with age, but there is less consensus on how this affects pricing, given the difficulty of accurately separating effects of cohort sizes from other economic conditions. We contribute to this literature in two important ways. First, our empirical strategy using lagged birth rates in a long-term historical context helps overcome many identification challenges. Second, we show that age-dependent entry into and exit out of homeownership is a key driver of developments in house prices and rent-price ratios, whereas earlier literature focuses on the price effects of general life-cycle patterns in housing consumption. These appear much milder.

We argue that the strong response of house prices and the much weaker effect on rent prices are related to market segmentation and the inelasticity of landlords and other market participants to changes in demand for homeownership coming from demographics. If a large cohort reaches their late 20s and starts desiring owner-occupied property, this cohort could significantly bid up prices for available properties if other market participants do not put more properties up for sale. As a result, house prices will increase relative to rents and some households will continue renting until supply increases and prices

[3] Leombroni et al. (2020) including housing in their study, which aims to explain the link between inflation, asset prices, and the entry of baby boomers to these markets during the 1970s. In their model, house prices equate the present value of future rents, while our empirical work aims to study how demographic demand for homeownership can change the relationship between house prices and rents.

fall again. We provide reduced-form evidence for such effects, but the role of segmentation has also been highlighted in theoretical work. In Greenwald and Guren (2021), shocks to the supply of credit can have large impacts on house prices and rent-price ratios if deep-pocketed investors do not respond to these shocks by putting more rental properties up for sale. Their calibration points to sizeable market segmentation and thus a large impact of demand shocks on house prices and rent-price ratios. Mabille (2023) studies young homebuyers and finds that changing credit constraints can lead to large and spatially heterogeneous changes in homeownership rates across cohorts. He further finds that the response of house prices and rents to shocks differs because of segmentation.

This paper proceeds as follows. Section I describes the demographic and housing data and the historical context. Section II presents stylized facts about demographics and the age-dependence of demand for rental and owner-occupied housing. Section III reports the main empirical results on the impact of lagged birth rates on the rent-price ratio. Section IV analyzes potential mechanisms driving our main results. Section V concludes.

I. Data and Historical Context

For the analysis in the paper, we rely on a large set of demographic and housing data from Amsterdam (predominantly) and Paris, spanning from the 16^{th} to the 19^{th} century. These data come directly from archives or are based on computations from existing historical studies. We introduce the key data sources in this section. Internet Appendix Table IA.V lists all historical data sources used in the paper.

In terms of housing market data, we rely on several sources. For transaction prices between the 1600s and 1810, we employ mandatory transaction registrations recorded by the Amsterdam aldermen. This data set includes all housing transactions in Amsterdam between 1700 and 1810 and most housing transactions from the 17^{th} century. We combine this database with 19^{th} century data on repeat sales on the Herengracht, a famous canal in Amsterdam, and with data from auction sales from other streets to estimate a continuous Amsterdam-wide repeat-sales index using Bayesian methods (Francke (2010)), which we use in this paper and has been presented in Korevaar, Francke, and Eichholtz (2021). Using the same method, we estimate additional house price indices for high- and low-homeownership areas between 1630 and 1810. To identify these areas, we make use of the 1805 rent census that provides rents and the homeownership rate for 75% of the Amsterdam housing stock. For rent prices in Amsterdam and Paris, we make use of the repeat-rent indices reported in Eichholtz, Korevaar, and Lindenthal (2020), who construct the indices using the same method as the house price indices and rely on new rental contracts only.

We combine archival sources and published studies to construct annual time series of demographic variables in Amsterdam and Paris. The key variable of interest is the birth rate. Time series of mortality, migration, and marriage rates are used as control variables in our main analysis. For Amsterdam, we also employ marriage data to link the age of a person to their housing market activity.

For Amsterdam, we use archival civil registers on all individual births, marriages, and deaths in the period before 1810, which have been digitized by the Amsterdam City Archives (ACA), available from 1554 (deaths) and 1565 (births, marriages). Records of births and deaths are organized by parish or cemetery, with some records missing, particularly in the early part of the sample. In those periods, we estimate demographic rates based on data from parishes that are consistently available. This design is comparable to that in other studies (e.g., Curtis (2016)) and is explained in more detail in Section I of the Internet Appendix. From the genealogy website *WieWasWie.nl*, we also obtain counts of births for several other cities.

After 1810, aggregate numbers of births, deaths, and marriages come from official statistics from the Amsterdam municipality. These sources also provide population and migration estimates from 1850 onward. For the period before 1850, we use estimates from historians available at five-year intervals. To transform these values to annual numbers, we use actual birth and mortality data, while interpolating the implied migration estimates from the population series. Since population series may contain errors, we construct a normalized migration series based on purchases of citizenship and registration in the protestant church, which we explain in Section I of the Internet Appendix.

For Parisian demographic data, we rely on a multitude of publications on annual numbers of births, deaths, and marriages from historians as well as official statistics from the city of Paris. We transform these into crude birth, marriage, and death rates based on the population numbers of Biraben and Blanchet (1999) for the prerevolution period and data from Paris statistical office thereafter. For Paris, we have no information on migration.

For supplementary analyses, and to obtain a set of controls, we rely on several other series. First, we obtain data on consumer prices and day wages in these cities since the 16th century. For Amsterdam, we also obtain data on GDP per capita and interest rates for the Province of Holland (pre-1800) and the entire Netherlands (post-1800). Interest rates are for perpetual government debt. To compare to stock prices, we use the series of dividend yields for the Dutch East India Company from Golez and Koudijs (2018), the main asset on the Amsterdam equity market in the period we study, covering 1629 until 1782. We also use a monetary measure of housing quality growth from Eichholtz, Korevaar, and Lindenthal (2020), which reflects the difference between average rent prices and the quality-controlled rent price index.

Beyond the historical data, we make use of OECD data on house prices, rent prices, and various demographic and economic factors for a panel of OECD economies from the 1970s until today. We use these data in Section VI of the Internet Appendix to examine the extent to which our historical findings might generalize to the present.

Descriptive statistics on the key variables introduced in this section are reported in Table I. As we restrict our analysis to the period before the

Table I

Descriptive Statistics, Annual

Economic variables are log differences, demographic variables are rates (0.01 = 1% rate). All series are continuous and end in 1884 for Amsterdam and in 1831 for Paris. The starting years differ across variables due to data availability. Note that the difference in mean rent price growth and house price growth is driven by the fact that the rent-price index covers a longer period. Over the same period, the difference is only 0.1%.

Statistic	Symbol	N	Mean	St. Dev.	Min	Max
			Amsterdam, 1550–1884			
Rental growth (log)	Δr	334	0.007	0.041	−0.191	0.278
House price growth (log)	Δh	264	0.004	0.051	−0.193	0.191
Wage growth (log)	Δw	334	0.006	0.014	−0.063	0.072
Inflation (log)	Δp	334	0.004	0.079	−0.258	0.337
Mortality	M	331	0.036	0.011	0.016	0.123
Nuptiality	N	320	0.010	0.002	0.006	0.020
Birth rate	B	320	0.034	0.004	0.019	0.045
Migration (Impl.)	Mi	320	0.009	0.018	−0.016	0.110
Migration (Norm.)	MiN	321	−0.049	0.979	−2.567	4.713
Population growth (log)	pop	334	0.008	0.016	−0.016	0.095
Δ Bond yield (log)	Δi	288	−0.003	0.087	−0.405	0.463
Δ Dividend yield	Δd	153	−0.001	0.042	−0.135	0.138
			Paris, 1500–1831			
Rental growth (log)	Δr	331	0.010	0.064	−0.475	0.434
Wage growth (log)	Δw	331	0.009	0.058	−0.223	0.223
Inflation (log)	Δp	331	0.009	0.074	−0.264	0.411
Mortality	M	287	0.036	0.011	0.012	0.153
Nuptiality	N	312	0.007	0.002	0.001	0.017
Birth rate	B	306	0.034	0.005	0.012	0.045
Population growth (log)	Pop	331	0.004	0.027	−0.260	0.174

demographic transaction, we provide descriptive statistics only for this period. As a conservative cutoff for the start of the demographic transition, we take the final birth rate peak before birth rates started trending down structurally, which happens in Paris in 1831 and in Amsterdam in 1884. The statistics on rent prices, house prices, wages, and consumer prices all refer to annual log-differences. Note that inflation was limited in this period, except in the late 16$^{\text{th}}$ century. Mortality, nuptiality (marriage rate), birth rates, and implied migration reflect percentage shares of the total population. For reference, we also include statistics on annual population changes in the table, but we should note that these are based on interpolated estimates from various studies, and thus their annual standard deviation likely underestimates the true standard deviation.

A. Institutional Context

To understand the data presented here and interpret our subsequent analysis, we describe some institutional features of the historical Amsterdam housing market, the primary focus of our study. Over the extensive time span of our research, key aspects of housing markets, such as credit availability, homeownership rates, and turnover rates, have changed significantly. We describe the most important characteristics of the market here and compare them with the modern context.

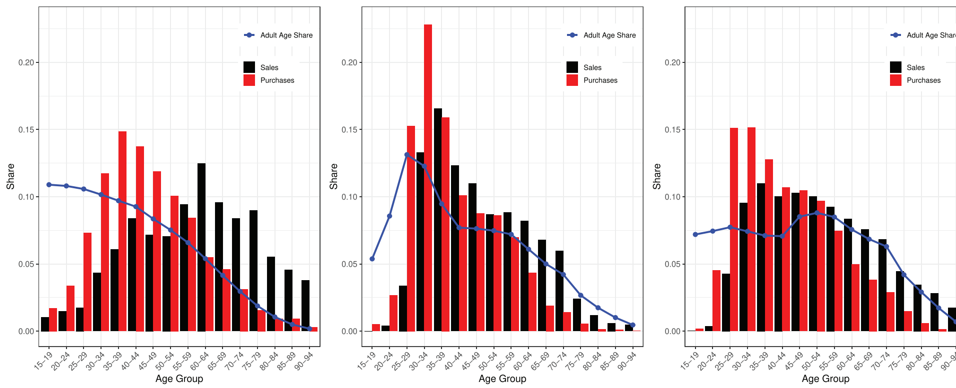
Historically, most real estate was owned by private individuals and could be traded freely, with transaction costs limited to a stamp duty averaging 2.5%, comparable to modern rates in the Netherlands. Other costs (broker, notary, etc.) amounted to around 1.5% of asset value (van Bochove, Boerner, and Quint, 2017). These costs are similar to those in modern-day Netherlands and are on the lower end compared to the United States. Although we do not know the exact extent to which real estate remained within families across generations through inheritance, it appears that many heirs sold inherited property, with approximately half of sales in the raw transactions data executed by heirs.

Commercial and institutional ownership of property was limited until the late 19th century. Overall, investment in housing accounted for about 30% of total household wealth (Korevaar, 2023). The extent to which properties were owner-occupied or renter-occupied varied over time. In the 16th century, about 41% of properties were owner-occupied (van Tussenbroek, 2019), but according to the rent register of 1805, only 29% of properties were occupied by their owners. As many properties were subdivided into smaller units such as rooms, basement units, and back houses, the rate of owner-occupancy at the housing unit level was just 14%. When interpreting this figure, one should consider that Amsterdam's low life expectancy resulted in a very young population. Young individuals typically moved out of their parental homes into rooms or other low-cost housing by their late teens, and only turned to homeownership or more independent rental housing later in life.

Homeownership rates likely reached their lowest levels during the 20th century, which we exclude from the analysis. During this period, Amsterdam developed a substantial rent-regulated social housing stock. More recently, homeownership rates increased from 6% in 1980 to about 30% today (Gemeente Amsterdam, 2019). Beyond the presence of social housing, the low rate of homeownership relative to the national average (60%) is likely due to Amsterdam being a large and expensive city, attracting young people who are more likely to rent, similar to many other global cities.

Recent growth in homeownership coincided with the rapid expansion of mortgage credit to households. However, for most of the sample period, mortgage provision was more limited, although it varied over time. During the 17th century, the number of annual long-term mortgages equaled about a third of the total number of housing transactions (Korevaar, 2023), with loan-to-value ratios reaching up to 100%. Credit during this period primarily came from peer-to-peer loans, a market that declined significantly in the 18th and early 19th centuries, before bank credit became dominant.

Although transaction activity in the housing market has always been cyclical, liquidity today seems quite similar to the historical average. During the 17th and 18th centuries, about 3% of the housing stock traded hands each



Panel A: Netherlands, 2019 Panel B: Amsterdam, 2019 Panel C: Amsterdam, 17th-18th century

Figure 1. Sales and purchases by age. This figure plots the distribution of buyer and seller ages in the Netherlands (Panel A) and Amsterdam (Panel B) in 2019, and in 17th to 18th century Amsterdam (Panel C). The historical data (C) include sales by heirs. This is estimated by linking repeat-sales pairs to the individual's marriage act, under the condition that there is a unique match with a high probability. The data are presented per five-year bin, and for reference we also provide the share of the adult population in each five-year bin. (Color figure can be viewed at wileyonlinelibrary.com)

year, whereas statistics from the Dutch Land Registry show this has been on average about 3.5% in the past 10 years. Less is known about the liquidity of the rental market. Surviving rental contracts from the archives indicate that agreements were typically for one year (e.g., Eichholtz, Korevaar, and Lindenthal, 2020). This is similar to the U.S. housing market today, whereas in modern Amsterdam, tenant protection laws have made permanent contracts the norm. To facilitate moving, contracts would typically start and end on May 1. Unlike most modern markets, there were no fiscal differences between rental property and owner-occupied property, which might have reduced frictions between renting and owning.

II. Stylized Facts

To motivate the main empirical analysis of the paper, we present three key stylized facts about demographics and the age-dependence of demand for rental and owner-occupied housing.

Stylized fact 1: *Both today and in history, individuals are net homebuyers when they are young adults and net sellers when they are old. Purchases and sales are correlated with cohort size.*

Figure 1 shows the age distribution of buyers and sellers of all properties sold in the Netherlands (Panel A) and Amsterdam (Panel B) in 2019, based on data from the Dutch Land Registry, as well as estimates for 17th to 18th century Amsterdam (Panel C). The latter are based on a small set of 967 repeat sales for which we could identify a unique link to the marriage act of the person buying and selling, and for which the ages of the bride and groom were recorded. The matching procedure is described in Section II.A of the Internet Appendix, where we also provide evidence for a similar age pattern in purchases and sales in modern United States. In blue, we plot a line with the share of the population over age 15 in each five-year bin based on actual data from modern Netherlands (Statistics Netherlands, 2023) and estimates for historical Amsterdam from van Leeuwen and Oeppen (1993).

In net terms, we observe a peak in net housing purchases when individuals are in their 20s and 30s. In modern Netherlands and Amsterdam, the peak occurs earlier compared to historical Amsterdam. In all three cases, the largest increase in net housing purchases is in the age 25-to-29 bin relative to the age 20-to-24 bin, although it is less pronounced in historical Amsterdam. Similarly, we observe a peak in net housing sales when individuals are in their 60s, with the peak occurring several years earlier in modern Amsterdam compared to the historical context. In all three cases, after net housing sales turn positive, we find the biggest increase when comparing the age 60-to-64 bin with the age 55-to-59 bin. This age pattern aligns with the idea that people become homeowners when they are young and move out of homeownership when they are old, either because they die or because they transition to other forms of housing as they age.

There are also important differences in the age patterns, for at least three reasons. First, buying a home requires capital, and in markets where homes are more expensive or mortgage provision is less generous, it might take prospective homebuyers longer to be able to purchase property. Modern Dutch first-time buyers are young, even compared to the United States, because the Netherlands allows homebuyers to take a 100% loan-to-value mortgage as long as they meet standard payment-to-income constraints. Historically, mortgage credit was more limited. In addition, because Amsterdam is an expensive city, first-time buyers today are older compared to the national average as it takes them longer to accumulate sufficient income (and savings) to buy there.

Second, there are differences in ownership structures across contexts. Similar to many other global cities, homeownership rates in Amsterdam are low relative to the national average, implying that many buyers and sellers are landlords. However, most rental property today is owned by nonprofit housing associations or commercial legal entities, implying that the age of the owner is not defined when these buy or sell. Historically, most rental property was owned by private individuals, for which we do observe the age. As landlords typically acquire property when they are older, this skews the distribution in historical Amsterdam toward older buyers and sellers. As we do not observe whether the owner ends up living in the property, we cannot adjust the historical data for this. Moreover, if heirs sold property in historical Amsterdam, the deceased owner would still be listed as the seller, often with the note "heirs of." Heirs did not always sell inherited property immediately after receiving it, which leads to a higher "age" at sale.

Third, if a particular cohort is large, we would expect more buyers and sellers from that cohort. Nowadays, for example, many young people move to

Amsterdam to study or work, but when they reach their 30s, they leave the city and start commuting to work from the suburbs. As we see in Figure 1, Panel B, this leads to a large spike in the number of young people and the number of properties they buy and sell. While Amsterdam has always been a city of migrants, this pattern is less evident in the demographic structure historically, as cars and rapid transit had not yet been developed, causing migrants to remain in the city proper until old age. Beyond migration, variation in cohort size can also be driven by differences in fertility and life expectancy, which we examine in the final stylized fact.

In the analysis, we exploit variation in cohort size based on differences in demographic rates. Because housing transactions are strongly age-concentrated, cohort differences in size could lead to sizable and predictable changes in demand for homeownership. Clearly, we observe in the data only the *realized* ages at which people buy and sell, not the ages at which individuals start desiring homeownership and at which they want to exit from it, either because they want to return to renting or they die. This brings us to the next stylized fact.

Stylized fact 2: *Households prefer renting in early adulthood and toward the end of their life cycle. As a result, demand for housing consumption is more stable across the life cycle than demand for housing investments*.

In the 2021 Dutch housing survey (Statistics Netherlands, 2022), 81% of adult household heads under 25 rent, which decreases to around 25% for household heads in their 40s and 50s. At older ages, households exit from homeownership because they pass away or because they turn to other forms of tenancy, such as specific property for seniors or living with their children. For Dutch household heads over 75, 43% rent. This pattern is also present in other countries.[4] The survey also asks residents about their preferred form of tenure. The majority of respondents under age 25 and over age 65 indicate a preference for renting, while the majority of respondents between ages 25 and 65 indicate a preference for owning.

These facts align with the idea that people prefer to rent by themselves when they are young and desire flexibility, and want to enter homeownership when they form long-term relationships and want to settle, most commonly in their late 20s. After retirement, people prefer renting again, for example, in old-age facilities. In Western Europe, this life-cycle pattern was already established in the premodern period (e.g., Hajnal, 2017). Children would typically leave their parental households in their late teens or early 20s, and marry in their 20s or early 30s. Historically, Amsterdam had a significant number of elderly homes, which were typically accessible for seniors aged 60 or older.

In short, demand for homeownership appears strongly age-concentrated among adults, while demand for housing consumption is much less so, both today and in history. This latter result also aligns with existing research, which establishes that housing consumption is negligible for children but rises significantly in early adulthood as children move out of their parental homes,

[4] See, for example, Engelhardt and Eriksen (2022) for patterns in old-age homeownership in the United States.

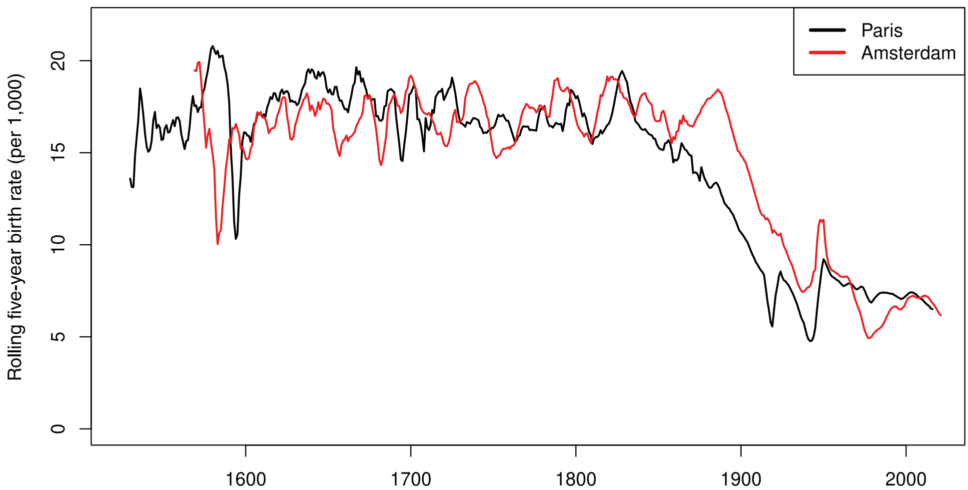


Figure 2. Birth rates. This figure plots the rolling estimated five-year birth rates per 1,000 inhabitants for both Amsterdam and Paris. Birth rates are high and volatile until the late 19th century, when they gradually start declining. Birth rates increase again after World War I (Paris) and after World War II (both cities). (Color figure can be viewed at wileyonlinelibrary.com)

and then stabilizes. Housing consumption is typically measured using the (rental) value of the property one lives in adjusted for the presence of other residents. While findings differ somewhat across studies about the exact age at which demand growth reduces or even turns negative, the overall pattern is quite consistent (e.g., Mankiw and Weil, 1989; Green and Hendershott, 1996; Eichholtz and Lindenthal, 2014). In Section II.B of the Internet Appendix, we show a similar pattern in the historical period.

The main takeaway is that changes in demographic rates might have very different impacts on the demand for rental housing relative to owner-occupied housing. In the owner-occupied market, only a small fraction of properties trades hands every year, and transaction activity is much more age-concentrated than in the rental market, where turnover is higher, and rental contracts are often renewed yearly or every few years. Whether this also leads to demographic shocks having a larger price impact in the owner market will ultimately depend on demand elasticities and the degree to which other investors react to shocks in demographic demand for homeownership, either by converting properties or by building more properties.

Stylized fact 3: *Fertility and mortality rates were high and volatile before the demographic transition, but declined and stabilized afterward. The economic impact of a given demographic shock will depend on the demographic regime*.

In Figure 2, we plot developments in total five-year birth rates from the 16th century until today for both Amsterdam and Paris, the key time series that we use in the main analysis of the paper. In the period before the demographic transition, birth rates were structurally high, although they fluctuated

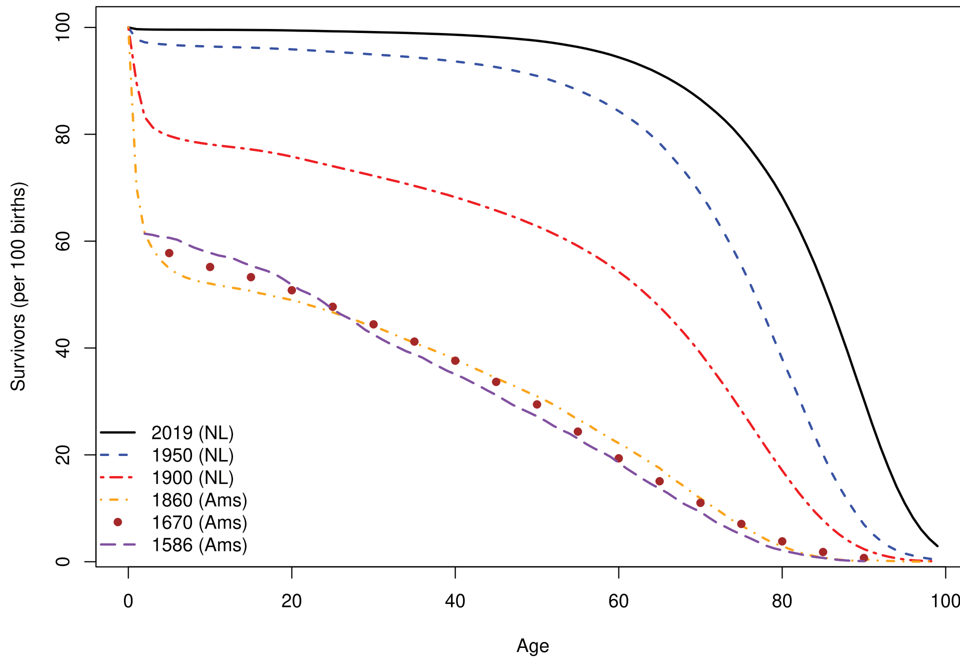


Figure 3. Survivalship rates. This figure plots the estimated survival rates per 100 newborns for Amsterdam (pre-1900) and the Netherlands (post-1900). (Color figure can be viewed at wiley-onlinelibrary.com)

substantially over time. There are about a dozen baby booms and busts in which five-year birth rates change by more than two percentage points.

In this analysis, we focus entirely on the period before the demographic transition. Due to the demographic transition, mortality and birth rates started trending downward substantially, only to be halted temporarily due to the postwar baby boom. An annual birth rate of 2% of the population (a five-year rate of 100 per 1,000) would be considered a large baby boom in modern terms, while it would be considered a large baby bust in historical terms. As a result, it is not possible to estimate the impact of demographic rates on the housing market in a single framework that uses data from both the modern and historical period. This problem is further exacerbated by the fact that the main demographic shifts around the World Wars coincided with major housing policy interventions, including strict rent controls and the development of large public housing systems.

The demographic transition also increased life expectancy. Figure 3 shows estimates of survival rates for Amsterdam and the Netherlands. Before the second half of the 19^{th} century, there was no structural change in survival rates and mortality rates. About half of newborns would not live until adulthood, and mortality remained significant even for younger adults. Afterward, life expectancy steadily improved.

These differences in life expectancy also imply that the meaning of a given birth rate shock (expressed as a fraction of the total population) differs across demographic regimes. In a society with high life expectancy, age cohorts are generally similarly sized until mortality increases at old ages, while in a society with low life expectancy, cohort size declines more linearly with age. As a result, a one-percentage-point birth rate shock will lead to a relatively large increase in the size of a birth cohort in a modern setting compared to the historical setting we study here. This also implies that the demand shocks that materialize when these cohorts reach adulthood or old age will be stronger in a modern context compared to a historical context. The effect will also be distributed differently: historically, mortality was less age-concentrated. This implies that there would be more limited concentration of sales at specific old ages and much less of an increase in rental demand from old-age own-to-rent transitions since few individuals would survive to such an age. Of course, the main mechanism behind this effect is the same in both low- and high-life expectancy regimes.

III. Analysis

We now aim to establish the relationship between changes in demographic structure and housing costs. The pronounced pattern in housing purchases and sales by age might imply that birth rates have a sizable effect on house prices, separate from their effect on rents, which reflects demand for housing consumption more broadly. The key challenge in identifying these relationships is that the demographic structure might be endogenous, particularly at the city level. Most notably, economically successful cities also tend to attract large inflows of migrants, which typically are relatively young adults. This is what we clearly observe in Figure 1, Panel B for modern Amsterdam.

Our main methodology examines the extent to which past shocks to birth rates explain changes in current rent-price ratios, as well as rent and house price growth individually. Our intuition is that a large number of babies today implies there will be a large cohort of x-year-olds x years from now, and that the size of past birth cohorts is a much more plausible exogenous shifter of housing demand conditions than current demographic structure. Before presenting the results, we describe the empirical model and address potential identification challenges.

A. Identification

To link changes in demographic rates to changes in demand for housing and homeownership, we need to link the size of cohorts to their demand. In the most basic model, demand for housing and homeownership is a direct function of age. In the early literature, Mankiw and Weil (1989) examine how housing consumption depends on residents' age by running a cross-sectional regression of individual house prices H_i on resident age fixed effects, for a given year, provided by $H_i = \sum_j \alpha_j \mathbb{1}(\text{age}_i = j) + \varepsilon_i$. Total housing demand in year t is then defined as $D_t = \sum_j \hat{\alpha}_j N_{jt}$, where N_{jt} is the number of people at age j in year t, assuming the weights vector $\hat{\alpha}$ is constant over time. Finally, the impact of housing demand on house prices is examined by running a time-series regression of aggregate log house prices or indices h_t on the estimated housing demand variable, $h_t = \delta_0 + D_t\,\delta_1 + \epsilon_t$, suppressing the dependence on control variables.

We cannot replicate this approach as we do not observe N_{jt} in our historical setting, apart from in 1851 (see Section II.B of the Internet Appendix). Empirically, the equation is typically estimated in differences as the level of house prices and demand are nonstationary. Because changes in the population and its housing demand are ultimately driven by changes in the size of age cohorts, for which past birth rates are a useful proxy, we assume that the change in demand D_t is a linear function of lagged birth rates $B_{t-\text{lag}}$: $\Delta D_t = \tilde{\alpha} + \sum_{\text{lag}} B_{t-\text{lag}}\,\tilde{\beta}_{\text{lag}}$. Substituting this relation in the aggregate house price equation (in changes) gives $\Delta h_t = \delta_1(\tilde{\alpha} + \sum_{\text{lag}} B_{t-\text{lag}}\,\tilde{\beta}_{\text{lag}}) + \varepsilon_t = \alpha + \sum_{\text{lag}} B_{t-\text{lag}}\,\beta_{\text{lag}} + \epsilon_t$.

In addition to Mankiw and Weil (1989), we estimate the impact of age cohorts or lagged birth rates on not only house prices, but also rents and rent-price ratios to be able to distinguish price effects resulting from shifts in ownership and consumption demand. In our empirical approach, we use changes over five years instead of annual changes, and the total birth rate over five years $B_{t-\text{lag}/4+\text{lag}} = \sum_{\tau=t-4}^{t} B_{\tau-\text{lag}}$, to estimate the impact of the demographic structure on prices and rents:

$$y_t - y_{t-5} = \alpha + \sum_{\text{lag}\in L} B_{t-\text{lag}/4+\text{lag}}\,\beta_{\text{lag}} + z_t'\gamma + \varepsilon_t, \tag{1}$$

where y is a vector of log rent price indices r, log house price indices h, or log rent-price ratios $r - h$, z_t is a vector of control variables, and L is a set of lags. We use five-year log changes to filter out noise and because changes in demographic structure are slow. A five-year period draws a good balance between picking up actual changes in demographic structure and statistical power.[5] We account for serial correlation from overlapping observations by computing Newey-West standard errors using a lag length of five (as we take five-year differences). We use these in all other estimation results we present in this section.

The minimum and maximum lag lengths in L we are considering are 15 and 70 years. Five-year birth rates 15 years ago correspond to birth cohorts that are between 15 and 19 years old. We start at 15 years old, since housing demand is negligible prior to that age. We do not extend beyond 70 to 74 years old because the share of teens that would reach this age was generally low and realized sales drop significantly too. We estimate equation (1) for different sets of lags L, ranging from including only one lag with various lengths, two lags (one for younger and one for older cohorts), to all lags between 15 and 70 years of age. We expect the coefficients to be large in absolute terms for lags

[5] Our results are not sensitive to the specific horizon: We obtain similar results when using 3- or 10-year changes.

corresponding to ages for which the *change* in housing demand is large, when young cohorts start buying houses, or older cohorts start selling their houses. Note that these lags do not need to be correspond to ages for which the level of housing consumption is highest or lowest.

We use changes rather than levels for all dependent variables and our nondemographic control variables because we could not reject the presence of a unit root.[6] This reduces the sensitivity of the rent-price ratio to measurement error, which could compound over time when using levels. All economic variables use nominal changes as structural inflation was limited in this period and we control for it directly.

Before getting to the results and their interpretation, we discuss the plausibility of our key identifying assumption that birth rates 15 to 75 years ago are exogenous to our set of dependent variables, conditional on a set of controls z_t. There are two key endogeneity concerns we have to address. The first is whether the arrival or removal of a large birth cohort from the housing market might coincide with other economic changes that affect prices. For example, a large birth cohort reaching peak home-buying age might put further upward pressure on prices if the presence of a large cohort of productive workers pushes up economic growth. In our analysis, we focus predominantly on the impact of lagged birth rates on rent-price ratios, as it is less obvious why any such effects would have a larger impact on rents or house prices. To also address this concern directly, we estimate the sensitivity of the main coefficients of interest to including contemporaneous demographic and economic control variables. We look at (five-year) birth rates (B), nuptiality (N), mortality (M), migration (Mi), and five-year log changes in wages (w), consumer prices (p), and GDP per capita (gdp).

The second concern is whether factors influencing birth rates today might also affect housing markets decades later, implying that the effect we study does not operate merely through birth rates but also through other factors. Again, this is more of a concern for effects on rent or house prices individually than their relative developments.

First, if birth rate shocks were highly local, some of the short-term pressure on the housing market could have been alleviated through migration. However, shocks in annual birth rates were highly correlated across cities in Holland, implying that they were regional. We establish this relation in Section III of the Internet Appendix, building on data on baptisms in nine other cities.

Second, annual birth rates in the predemographic transition period covaried with climatological factors (most notably temperature), economic factors (most notably inflation), and demographic factors (predominantly marriage and mortality rates). In Internet Appendix Table IA.III, we show full regression

[6] We tested for the null of a unit root using Augmented Dickey-Fuller (ADF) tests, Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests, and Dickey-Fuller Generalized Least Squares (DF-GLS) tests. In most cases, all tests give the same results; we take first differences if one of the tests indicates the presence of a unit root. All variables are stationary in first differences.

output linking birth rates in Paris, Amsterdam, and other Dutch cities to the set of variables for which we have data. Overall, these factors are largely in line with existing literature on the drivers of birth rates in the premodern period (e.g., Wrigley and Schofield, 1981; Galloway, 1988; Bailey and Chambers, 1998; Crafts and Mills, 2009; Waldinger, 2022). While it seems intuitively implausible that shifts in these variables still affect outcomes decades later, we attempt to control for such concerns directly by also including lags of our set of control variables that match the birth rate lags ($B_{t-\mathrm{lag}/4+\mathrm{lag}}$).

B. Results

In Table II, we present our main results using five-year changes in rent-price ratios as the dependent variable. We report the results for the two main lags of five-year birth rates: births 25 to 29 years ago and 60 to 64 years ago. These correspond to the main periods in which people on average marry and start to consider moving to owner-occupied housing, and the ages at which they die or move back to rental housing. If such transitions significantly change the demand for owner-occupied housing relative to rental housing, we could expect particularly important changes in rent-price ratios at these lags. We find large price effects at these lags, but we also report on other lags below. A specification with all birth rate lags in a single regression is reported in Internet Appendix Table IA.VI.

In column (1), we present results for our baseline predictive regression without any controls. We find that a one-percentage-point increase in the current five-year birth rate predicts around a 4% decline in rent-price ratios 25 to 29 years later and a 2.8% increase 60 to 64 years later. Taken together, these two birth rate lags alone explain 18.4% of the variation in rent-price ratios over a long time span of over 250 years.

In column (2), we add demographic controls. Not surprisingly, we find a positive relation between death rates and a negative relation with migration. Note that because mortality was high among children and also common at younger ages, birth rates 60 to 64 years ago might still be a better proxy for old-age mortality than the current death rates, whose variation comes mostly from large shocks such as epidemics. Overall, adding demographic controls does not change the results on our main coefficients of interest, and if anything makes them somewhat stronger. In column (3), we add economic controls for wage growth, inflation, and changes in GDP per capita. Again, this does not result in sizable changes in our coefficients of interest. Column (3) is our baseline specification including results in the remainder of the paper. For robustness, however, we also consider alternative specifications.

In column (4), we add lagged demographic and economic controls corresponding to the lags of the two birth rate lags, to address concerns that changes in birth rates might correlate with other economic or demographic changes that could still affect outcomes decades later. Although this significantly increases the number of parameters to be estimated, it does not seem to alter our results. In the final two columns, we look at the effect of two controls

Table II

The Effect of Lagged Birth Rates on Rent-Price Ratios in Amsterdam

This table provides estimation results from equation (1) for rent-price ratios $(r - h)$ on lagged birth rates (B) in Amsterdam. Other variables are explained in Table I. The number of observations is 256. We report Newey-West standard errors using a lag length of five. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Dependent Variable:	$(r-h)_t - (r-h)_{t-5}$					
	(1)	(2)	(3)	(4)	(5)	(6)
$B_{t-25/29}$	-3.952^{***}	-4.353^{***}	-4.251^{***}	-5.019^{***}	-4.441^{***}	-4.775^{***}
	(1.122)	(1.139)	(1.130)	(1.081)	(1.109)	(1.015)
$B_{t-60/64}$	2.803^{**}	3.947^{***}	4.001^{***}	3.172^{***}	4.120^{***}	3.472^{***}
	(1.081)	(0.855)	(0.837)	(1.065)	(0.789)	(0.658)
$B_{t-0/4}$		0.677	0.562	1.048	0.346	0.300
		(0.953)	(0.924)	(0.982)	(0.872)	(0.747)
$M_{t-0/4}$		1.465^{***}	1.143^{**}	0.687	1.213^{***}	0.918^{**}
		(0.441)	(0.442)	(0.597)	(0.413)	(0.374)
$N_{t-0/4}$		−2.203	−2.701	−2.394	−2.941	−2.777
		(2.143)	(2.155)	(2.269)	(2.022)	(1.924)
$Mi_{t-0/4}$		0.249	0.432	0.618	0.345	0.418^{*}
		(0.272)	(0.287)	(0.406)	(0.293)	(0.233)
$MiN_{t-0/4}$		-0.014^{***}	-0.014^{***}	-0.017^{***}	-0.014^{***}	-0.014^{***}
		(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
$w_t - w_{t-5}$			−0.487	-0.668^{*}	−0.426	−0.492
			(0.369)	(0.348)	(0.372)	(0.316)
$p_t - p_{t-5}$			0.129^{*}	0.206^{***}	0.113	0.051
			(0.069)	(0.062)	(0.075)	(0.064)
$gdp_t - gdp_{t-5}$			−0.079	−0.097	−0.075	−0.045
			(0.086)	(0.071)	(0.089)	(0.073)
$hq_t - hq_{t-5}$					0.276^{**}	
					(0.127)	
$i_t - i_{t-5}$						0.266^{***}
						(0.066)
Constant	0.200	−0.215	−0.135	0.076	−0.084	0.136
	(0.289)	(0.365)	(0.370)	(0.371)	(0.345)	(0.315)
Lagged controls	No	No	No	Yes	No	No
Adjusted R^2	0.178	0.433	0.448	0.520	0.465	0.554

that form a prelude to our later discussion of possible mechanisms. In column (5), we add a control for changes in average housing quality, which is defined the change in average rental prices relative to the change in the rent price index (Eichholtz, Korevaar, and Lindenthal, 2020). This reflects changes in the rental housing stock, primarily due to renovations or new construction. Note that this measure covaries positively with rent price growth because such construction generally occurs when housing consumption demand is high. This is why we do not include it in the baseline. We study the impact of lagged birth rates on construction in more detail in the next section. However, its inclusion does not alter our main results.

Table III

The Effect of Lagged Birth Rates on Rents and House Prices in Amsterdam and Paris

This table provides estimation results from equation (1) for rents (r) on lagged birth rates (B) in Amsterdam and Paris, and house prices (h) in Amsterdam. We report Newey-West standard errors using a lag length of five. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Dependent Variable:	$h_t - h_{t-5}$			$r_t - r_{t-5}$		
	Amsterdam					
	(1)	(2)	(3)	(4)	(5)	(6)
$B_{t-25/29}$	3.645***	4.089***	4.457***	−0.307	−0.162	−0.562
	(1.380)	(1.395)	(1.314)	(0.726)	(0.612)	(0.651)
$B_{t-60/64}$	−3.106*	−4.708***	−4.033***	−0.302	−0.707	−0.862*
	(1.629)	(1.126)	(1.377)	(0.668)	(0.465)	(0.509)
Observations	256	256	256	256	256	256
Adjusted R^2	0.122	0.493	0.587	−0.001	0.251	0.310
Paris						
				(7)	(8)	(9)
$B_{t-25/29}$				1.345	2.505**	−0.163
				(1.349)	(1.225)	(0.784)
$B_{t-60/64}$				1.906	0.329	−0.250
				(2.257)	(1.340)	(0.610)
Observations				242	242	217
Adjusted R^2				0.031	0.374	0.319
Controls						
Demographic	No	Yes	Yes	No	Yes	Yes
Economic	No	Yes	Yes	No	Yes	Yes
Lagged (both)	No	No	Yes	No	No	Yes

In column (6), we make a first attempt to investigate whether the effect on rent-price ratios is specific to housing markets or whether it reflects a general asset price effect by including a control for interest rates. Including this control leaves the main results unchanged, although increasing the negative effect on lags 25 to 29 years ago and decreasing the old-age effect somewhat. This suggests that the young-age effect in particular is housing-specific. We look at impact of lagged birth rates on other assets in the next section. In the baseline specification, we do not control for interest rates as both could be affected by lagged birth rates.

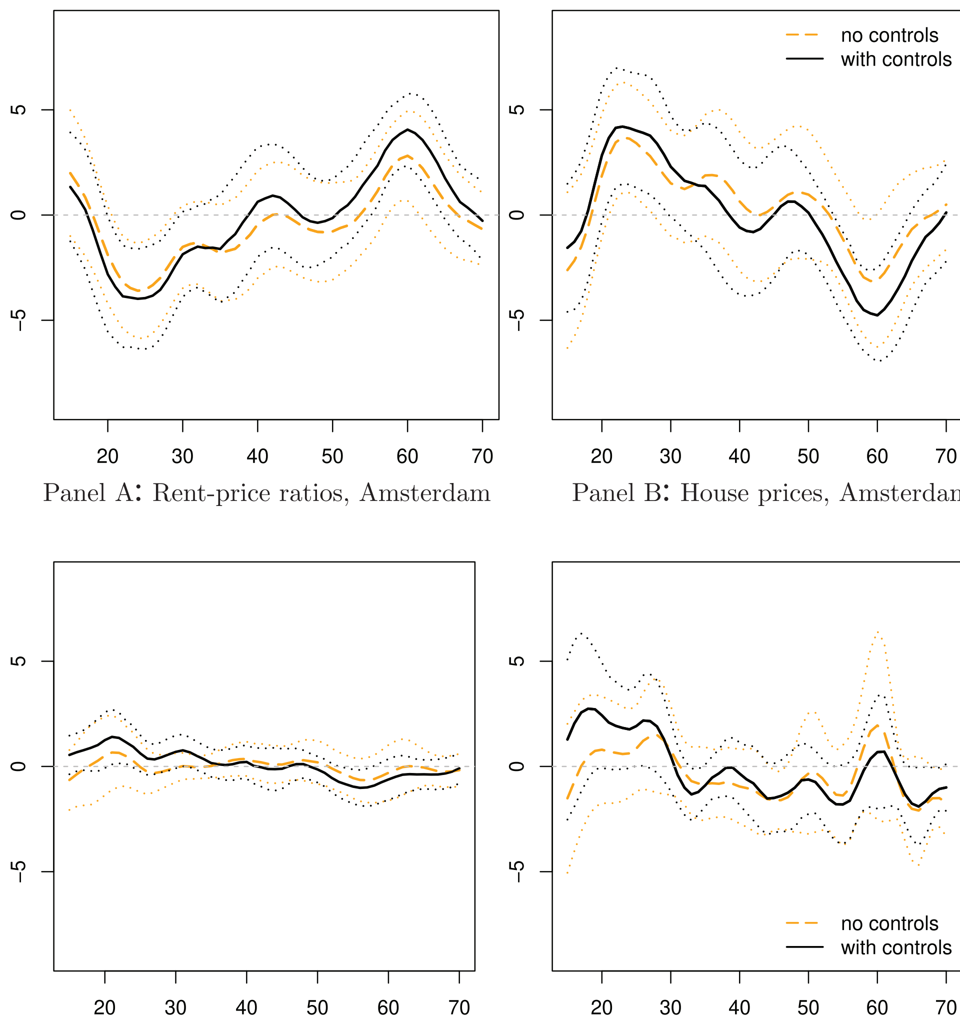
Is the effect that we document caused by changes in house prices or by changes in rents? To investigate this question, Table III takes a closer look at the effects of lagged birth rates on rent and house prices individually. In this table, the dependent variables are the five-year changes in the house price and rent index. For Amsterdam, we analyze the changes in both house prices and rents. We also look at rents in Paris. We report results without any controls (columns (1), (4), and (6)), with all contemporaneous controls (columns (2), (5), and (7)), and with all contemporaneous and lagged controls (columns (3), (6), and (9)).

We start with the effect on Amsterdam house prices in columns (1) to (3). Irrespective of the controls we include, we find a large positive effect of birth rates 25 to 29 years ago on house prices, and a large negative effect for birth rates 60 to 64 years ago. We next examine rents, holding the sample period fixed to facilitate comparison. For Amsterdam, we find no clear effects of lagged birth rates on rent price growth, not for the lags corresponding to young adults or those for older ages. The coefficients are also economically small. For Paris, we also do not find consistent effects, although the 25- to 29-year lag shows up significantly in the specification with contemporaneous controls. The lack of a rent price effect is not driven by the stickiness of rent prices. In modern Amsterdam, rent contracts are typically permanent, and for most properties rent levels and increases are capped by legislation, but these regulations did not exist in the period we study. Both the rent index for Amsterdam and that for Paris are based on newly signed contracts only.[7] The main conclusion is that the effect on changes in rent-price ratios appears to be driven predominantly by changes in house prices, with lagged birth rates at best having very weak effects on rental prices.

We now investigate how these results vary depending on the exact lag length we have chosen. The effects on prices generally concentrate at the 25- to 29-year lag and the 60- to 64-year lag. These lags also correspond to the period in which, in the transactions data, in both modern and historical times, we observe the largest increases in net housing purchases (25 to 29) and net housing sales (60 to 64), and the ages at which we see a significant switch in modern Dutch survey data on tenure preferences. Nonetheless, we do not rule out that demographics also play a role for other lags. To investigate this possibility, Figure 4 presents estimated lagged birth rate coefficients and corresponding 95% confidence intervals for different individual lag lengths and for each of the four dependent variables: Amsterdam growth in rent-price ratios, house prices, and rent prices, and Paris rent growth.[8] Each point on these lines represents a regression model with the birth rate lags varying across models. For clarity, a birth rate lag of 20 is the total five-year birth rate 20 to 24 years ago. We report estimates both without control variables, to test for pure time-series predictability of that particular lag, and with controls, to adjust for contemporaneous changes in economic and demographic variables that might also affect housing demand, in line with column (3) in Table II and columns (2), (4), and (6) in Table III. Note that we can use somewhat more

[7] Since rent contracts are unavailable for most Amsterdam data points, the index includes only cases in which rent prices changed compared to the previous year—this was only possible in the case of a contract change.

[8] As a robustness check, Figure IA.5 in the Internet Appendix reports coefficients and 95% confidence intervals based on the median standard error from five nonoverlapping regressions, where we shifted the start year by one year for each regression. This gives similar results.



Panel A: Rent-price ratios, Amsterdam

Panel B: House prices, Amsterdam

Panel C: Rent prices, Amsterdam

Panel D: Rent prices, Paris

Figure 4. The effect of lagged birth rates on rents, prices, and rent-price ratios. These figures report the elasticity (in %) of lagged five-year birth rates on five-year changes in rent-price ratios, and house and rent prices. For Amsterdam, we use transactions up to 1884, and for Paris up to 1831. Each point corresponds to a different regression, with the birth rate lag and starting year changing over regressions. The birth rate lag is reported on the x-axis. For example, a birth rate lag of 20 corresponds to the birth cohort currently aged 20 to 24. Birth rate data for Paris start in 1526 and for Amsterdam in 1565. Regressions include either no controls (full line) or contemporaneous + lagged economic and demographic controls; see column (3) of Table II. Dashed lines reflect 95% confidence intervals, based on Newey-West standard errors using a lag length of five years. (Color figure can be viewed at wileyonlinelibrary.com)

observations for rent prices when using lags corresponding to younger cohorts; the longer the lag, the more observations we lose.

Starting with the effect on rent-price ratios in Figure 4, Panel A, we find that the negative effect is largest for cohorts in their mid-20s and gradually disappears once we select cohorts in their 30s. For older cohorts, we see positive effects peak for the birth cohort in its early 60s. There are no significant effects for other cohorts. The effect on house prices, visible in Figure 4, Panel B, is almost the mirror image of the rent-price ratio effect, in line with the results in Table II.

The bottom two panels show the effect on rent prices in Amsterdam (left, Figure 4, Panel C) and Paris (right, Figure 4, Panel D). Without controls, none of these lags has a significant relationship to rent prices. Including controls, we find some weak positive effects for young cohorts and negative effects for older cohorts, although both economically and statistically much less significant than the effect on changes in house prices and rent-price ratios.

This also brings us to the key conclusion of this analysis: we find that sizes of birth cohorts positively predict house prices when this cohort reaches its mid-20s, while effects on rent prices are much weaker. At old ages, particularly when individuals hit their 60s, the effect reverses. As a result, the large effects that we find on rent-price ratios are driven predominantly by adjustment in house prices and not rents. While the timing of these effects coincides with shifts in the demand for homeownership, suggesting that such shifts do indeed have price effects, the price results themselves do not tell us why these price effects obtain. We investigate this question in Section IV.

C. Modern Context

To what extent do such effects hold in the modern context, given the large economic and demographic changes most countries have experienced since the 19th century? Data limitations and gradual demographic shifts prevent us from perfectly extrapolating our empirical approach to modern data, but we use a more limited and modified approach to test whether there is any evidence for similar relationships in a panel of modern OECD countries in the period from 1970 to 2020. The results, reported in Section VI of the Internet Appendix, show that in the modern context, we again see a clear pattern whereby increases in the share of young children predict prices and rent-price ratios but not rents. This suggests that our findings also generalize to the modern context.

IV. Mechanisms

In this section, we engage in additional empirical analysis to better understand the mechanisms behind our main result. While we have motivated our price results so far as being driven by strong age-dependence in demand for homeownership, we have not directly examined the extent to which lagged birth rates affect transaction activity and whether there is any evidence for segmentation between rental markets and owner-occupied markets, which could explain the differential response in house prices and rents. We address these issues in the first two subsections.

We next study whether there is any evidence that alternative mechanisms could play a role in the price effects we find. In particular, we study whether supply constraints contribute to the fact that house prices rise significantly in response to demographic demand for homeownership. We also study whether the effects we find are specific to housing, which would be in line with our main mechanism, or whether we find similar effects when looking at the yields of other assets. This would suggest that any price effects reflect broader age-dependence in demand for investment assets.

In deciding which exercises to conduct, we had to make a trade-off between the relevance and precision of each analysis on the one hand and the availability of data on the other. Our analysis requires supplementary long-term data, but is constrained by the materials that survive in archives and have been or could be digitized. In practice, this means we sometimes need to rely on proxies or on shorter samples. In most cases, our analysis is limited to the period before 1800, as most data sources stop during the period when Amsterdam was under French rule (1795 to 1814), while the period afterward is too short to investigate separately.

A. Probability of Sale

We start by directly examining the extent to which the arrival and departure of large birth cohorts in the housing market affects overall probabilities of the sale of existing properties. In our stylized facts, we document a pronounced age pattern in housing transactions, with people buying at young ages and selling at older ages, which likely relates to the fact that people want to enter homeownership when they start settling and exit homeownership when they are old or pass away. Landlords could respond to this demand by selling properties to prospective owner-occupiers when there is a large cohort of young adults desiring owner-occupancy. Conversely, landlords could purchase owner-occupied properties and convert them to rentals when there is a large older cohort that wants to exit homeownership. In short, the more responsive landlords or other market participants are to these shocks, the smaller the price effect this would generate. On average, about 3% of the housing stock traded hands each year, so the arrival (departure) of a large birth cohort could significantly increase (decrease) competition for properties for sale if there is only a limited response in transactions.

Ideally, we would like to estimate the extent to which lagged birth rates affect the probability of transitioning from rental to owner-occupied property, and vice versa. Unfortunately, we do not observe this in the data. However, using sales data, we can infer how likely a particular property is to sell. If increased demand for homeownership leads some incumbent owners to sell their property to first-time buyers, we would expect lagged birth rates to increase the likelihood that a property is sold. Similarly, we would expect a higher probability of sale if a large cohort reaches old age and starts selling their properties. In short, we expect higher transaction activity at young and old ages, and less in between.

To test this prediction, we use a discrete-time hazard model based on repeat-sales data for the period 1620 to 1811 from Korevaar (2023), covering the entire Amsterdam housing market. Conditional on the first sale within a repeat-sales pair, we model the probability that property i is sold after $t_i - s_i$ years, $\Pr(D_{is} = t_i - s_i)$, where s_i is the date (in years) of the first sale, t_i is the date of the second sale, and D_{is} is a stochastic variable of the holding period.[9] If the last sale of property i is before the sample end date, we have right-censored data for this last period. The probability that property i is not sold after $T - s_i$ years is given by $\Pr(D_{it} > T - s_i)$. These probabilities can be expressed as

$$\Pr(D_{is} = t_i - s_i) = H_{is}(t_i - s_i)S_i(t_i - s_i - 1), \tag{2a}$$

$$\Pr(D_{it} > T - s_i) = S_i(T - s_i), \tag{2b}$$

where $H_{is}(d) = \Pr(D_{is} = d | D_{is} > d - 1)$ is the hazard function or conditional probability, $S_{is}(d) = \Pr(D_{is} > d) = \Pi_{j=1}^{d}(1 - H_{is}(j))$ is the survival probability, and d is the duration. We specify the hazard by a sigmoid function, $H_{is}(d) = (1 + \exp(-x'_{isd}\beta))^{-1}$, so we can apply a logistic regression to estimate the discrete-time hazard model given by equation (2) in “property-period” format. The dependent variable is zero, except for the year of the second sale, when it is one (see Jenkins (1995) for more details). Figure 5 shows the coefficients for different lag lengths of the total five-year birth rate in the logit regressions, without and with control variables (similar to column (3) in Table II).[10]

We find that the coefficients on lagged total five-year birth rates are positive for lags between 25 and 40 years, negative for lags between 41 and 62 years, and positive again for larger lags. The sign of the coefficients and turning points are roughly similar when control variables are excluded or included.

The findings in Figure 5 indicate that cohort size is an important driver of turnover in the housing market. In line with Figure 1, large young and old cohorts lead to more transaction activity, while there is much less activity in between when cohorts are less likely to transition between owning and renting.

The arrival of a large birth cohort at purchasing age and their departure at older ages increases the probability of a sale. However, this effect occurs several years after we document price effects and is particularly slow to materialize for the arrival of a large birth cohort. The peak and trough for the price effect are at approximately lags 25 and 60, while the peaks for the probability of sale are later at lags 37 and 67. A one-percentage-point increase in the total five-year birth rate with a lag of 37 years raises the probability

[9] In the case of more than two sales per property, we create multiple repeat-sales pairs and treat them similarly.

[10] We report results for the two main lags of five-year birth rates, births 25 to 29 years ago and 60 to 64 years ago, in Internet Appendix Table IA.VII.

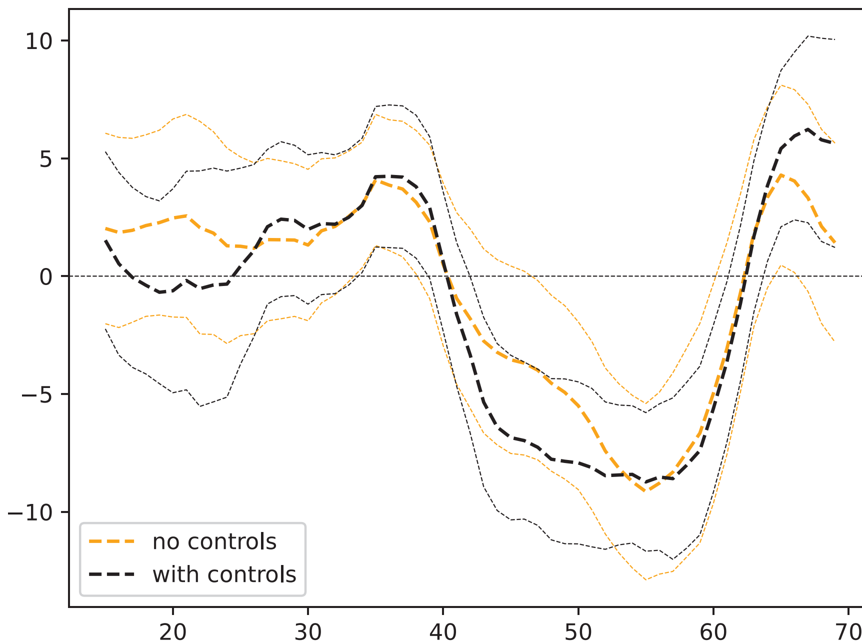


Figure 5. Coefficients of lagged birth rates in the probability of the sale model for Amsterdam. This figure plots the coefficients of lagged five-year birth rates in the probability of the sale equation, equation (2), for different lag lengths. Each point corresponds to a different regression, with the birth rate lag and starting year changing over regressions. The birth rate lag is reported on the x-axis. For example, a birth rate lag of 20 corresponds to the birth cohort currently aged 20 to 24. Controls include current demographic rates and changes in wages, consumer prices, and GDP per capita (see Internet Appendix Table IA.VII). Dashed lines reflect 95% confidence intervals, based on Newey-West standard errors using a lag length of five years. (Color figure can be viewed at wileyonlinelibrary.com)

of a sale by 0.1 percentage points, an increase of 4%. For the 67-year lag, the results are similar. Therefore, the economic impact on the probability of a sale is significant but relatively small at individual birth rate lags.

Overall, the results indicate that higher demographic demand does result in additional transactions, but this effect is more spread out and appears later than the effects in prices. While the data do not allow us to determine why the effect on turnover is more spread out, one possible explanation is that incumbent owners are not very elastic and become willing to sell a property to a large young cohort only after prices have increased significantly. Conversely, heirs or seniors selling property at old ages might hold on to their properties longer if they have to sell at a discount.

B. Segmentation of Markets

We have now documented that demographic structure, identified using variation in lagged birth rates, affects not only prices but also the number of transactions. The latter response appears to materialize more gradually. In this subsection, we examine whether there is any evidence that segmentation between rental and owner-occupied markets contributes to the price effects we find. If rental and owner-occupied markets are segmented, this might weaken the extent to which landlords convert rental property to owner-occupied property and vice versa.

The question is how to define market segments with the data we have available. We focus on market segmentation based on spatial differences in homeownership rates. Our intuition is that most urban housing markets have both areas predominantly catering to owner-occupiers and areas catering mostly to renters. When people reach the ages at which they start to demand homeownership, they might desire to move from one area to another. For example, two renters marrying in their mid-20s might have been living individually in rental units in their late teens and early 20s, and now look to buy owner-occupied housing together. This dynamic could place downward pressure on house prices in rental-dominated areas and upward pressure on house prices in areas with more owner-occupiers, where they dominate the market.

To test this conjecture, we use the same repeat-sales sample as in the previous subsection. We compute street-level homeownership rates based on the housing census of 1805, available from the ACA, and link them to repeat-sales pairs. Although only three quarters of the census survived, it is the only census that covers individual housing units rather than entire buildings, listing tenancy/owner-occupancy status. Note that the fact that we can observe owner-occupancy rates in a single year, not for the entire city, and only up to 1811, weakens our ability to identify any effects, and hence our results should be treated as suggestive.

We split the matched repeat-sales sample into three equal-sized parts: sales taking place on streets with low, medium, and high homeownership rates. Areas with low homeownership rates typically consist of streets with many cheap housing units in a single building (i.e., rooms, parts of properties). These streets typically have no or barely any owner-occupied properties, and an average annual rent of 50 guilders. High-homeownership-rate areas are found in the more expensive areas, such as the famous canal ring, with rents averaging over 300 guilders. Although most homes in these areas are owner-occupied, most have a portion, such as the basement or an outbuilding, that is rented out. In short, segmentation is related not only to rental versus owner-occupied housing, but also to the difference between cheaper and more expensive segments of the housing market.

We estimate a repeat-sales model with separate annual indexes for low-, medium-, and high-homeownership areas. Details are provided in Section V of the Internet Appendix. The estimated indexes are shown in Internet Appendix Figure IA.6. We only use the index from 1637 onward; before this period there are no prices of nonforeclosure sales, implying much uncertainty about the values of the triple index. Note that the overall index closely tracks the index that we use in the remainder of the paper, which employs all available sales. We then reestimate our main specification as given in

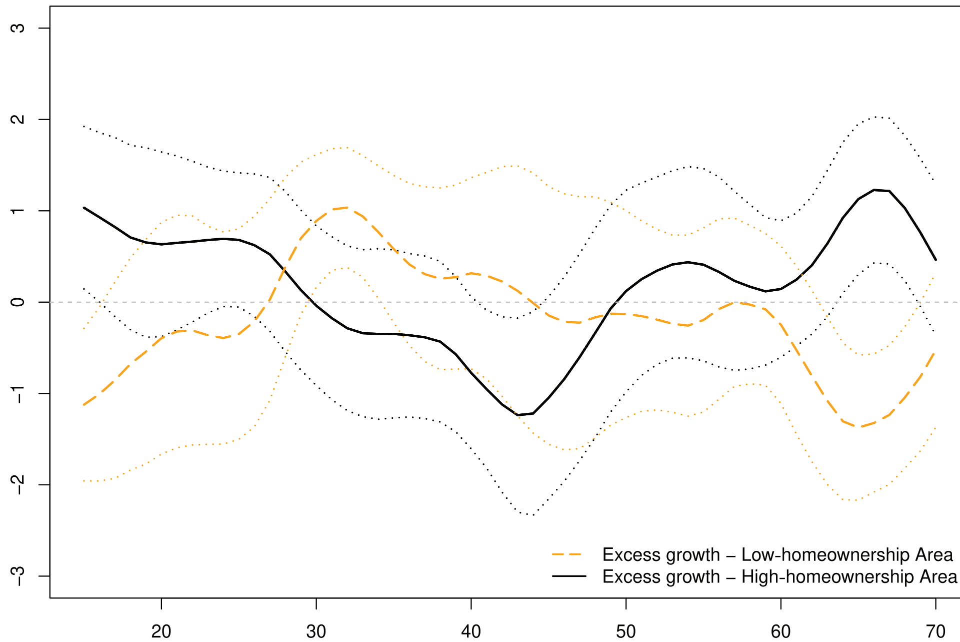


Figure 6. The effect of lagged birth rates on prices in high/low- homeownership areas. These figures report the effect of lagged five-year birth rates on five-year changes in house prices in high-homeownership and low-homeownership areas (solid line), relative to baseline in home prices in average districts. Each point corresponds to a different regression, with the birth rate lag and starting year changing over regressions. The birth rate lag is reported on the x-axis. For example, a birth rate lag of 20 corresponds to the birth cohort currently aged 20 to 24. Controls include current demographic rates and changes in wages, consumer prices, and GDP per capita; see column (3) of Table II. Dashed lines reflect 95% confidence intervals, based on Newey-West standard errors using a lag length of five years. (Color figure can be viewed at wileyonlinelibrary.com)

equation (1), using five-year changes in the low- and high-homeownership areas as dependent variables.

Figure 6 shows the coefficients for different lag lengths of the total five-year birth rate in the regressions including the standard set of control variables. We observe that for birth rate lags corresponding to teens, house price growth is higher in low- compared to high-homeownership areas. For a one-percentage-point increase in the birth rate, this results in about 2% higher house price growth in low-homeownership areas compared to high-homeownership areas. This aligns with observations that when large birth cohorts reach adolescence and begin moving out, this results in increased demand pressure predominantly in areas with low homeownership. These are also the areas where rooms are most commonly found.

As we move to larger lags of birth rates, these trends gradually begin to reverse. For lags corresponding to individuals in their early 30s, we begin to find statistically significant excess house price growth in areas of high homeownership, amounting to about 1%. The effect occurs somewhat later than the main effect on house prices and rent-price ratios. This is not surprising, since areas with the highest owner-occupancy rates typically consist of the most desirable properties that are usually purchased later in life. This might also explain the small negative effect on low homeownership areas for birth rate lags corresponding to cohorts in their 40s. Conditional on surviving to that age, one may have enough income and savings to buy a house somewhere more in demand.

Note that these effects should also be treated as suggestive, as the price difference between high- and low-homeownership areas cannot always be precisely identified for these cohorts. We observe a more consistent divergence in house price growth across low- and high-homeownership areas for birth rate lags corresponding to cohorts in their 60s. Prices decrease in areas with high homeownership rates and increase in areas with low homeownership rates. The difference is about 1% on both sides. The most likely explanation is that the heirs of homeowners start selling their assets either after death or because homeowners move back into rental housing.

In summary, the disparate house price effects of lagged birth rates on high- and low-homeownership areas, and the timing of these effects, align with the housing life cycle of renting early in life, moving to an owner-occupied property, and then moving back to renting or exiting the market at older ages. These price effects suggest some segmentation in the housing market, implying that house prices respond rather strongly to specific demographic demand shocks for rental or owner-occupied housing.

C. Construction and Supply Constraints

In the prior subsections, we examine price trends and sale probabilities for the existing housing stock, providing suggestive evidence that frictions in converting rental housing to owner-occupied housing and vice versa contribute to the price effects we find. While this is the most intuitive margin of adjustment, it is also possible that if demographic demand increases the demand for homeownership, developers respond by building additional housing, particularly if price increases push house prices above replacement costs.

In this section, we aim to examine whether supply constraints prevented such developments from happening. If that were the case, the price effects that we find in response to construction may be a mechanical product of supply constraints, at least in part. However, this would still not explain why prices move more than rents. When it comes to actual construction, Section IV of the Internet Appendix uses several sources to develop proxies for construction over the 17th to 19th centuries and finds some evidence that large young cohorts correlate with additional construction while older cohorts correlate with less new-construction activity.

In Amsterdam, the urban government has always taken a highly planned approach to urban expansion, with its monumental canal ring a living testament of this approach. When the population was growing rapidly, the city would develop plans to expand that were complex, costly, and typically took

Table IV

Constrained Supply and Price Effects

This table provides estimation results from equation (1) for rent-price ratios $(r - h)$ on lagged birth rates (B), adding an interaction term with periods of urban growth and correspondingly sizeable supply constraints (SC) in columns (2) and (4). The supply-constrained dummy is equal to one before 1668 and after 1855. Columns (3) and (4) including contemporaneous controls for demographics and economic conditions. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Dependent Variable:	$(r-h)_t - (r-h)_{t-5}$			
	(1)	(2)	(3)	(4)
$B_{t-25/29}$	-3.952^{***}	-4.104^{***}	-4.251^{***}	-4.778^{***}
	(1.122)	(1.285)	(1.130)	(1.292)
$B_{t-60/64}$	2.803^{**}	3.058^{*}	4.001^{***}	4.047^{**}
	(1.081)	(1.781)	(0.837)	(1.637)
SC		0.023		−0.758
		(0.613)		(0.560)
$SC \times B_{t-25/29}$		0.530		4.188
		(2.374)		(2.735)
$SC \times B_{t-60/64}$		−1.089		0.136
		(2.122)		(1.715)
Constant	Yes	Yes	Yes	Yes
Demographic controls	No	No	Yes	Yes
Economic controls	No	No	Yes	Yes
Observations	256	256	256	256
R^2	0.184	0.233	0.469	0.478

years to decades to materialize (see Abrahamse, 2010; Smid, 2019). As a result, housing supply was significantly constrained in periods when the city was growing rapidly and expansion plans had not been completed. Amsterdam experienced two such episodes in the period we study: first during the Golden Age until 1668, and again after 1855 when population growth picked up following industrialization.

On the other hand, there were few restrictions on the expansion of housing supply as long as vacant lots within the city walls were available for (re)development. Importantly, it was the slow expansion itself that gave rise to this situation. Amsterdam started making plans for a fourth expansion of the city in the 1650s based on the growth it had been experiencing at the time (see Abrahamse, 2010). However, by the time the plans were realized in the 1660s and 1670s, population growth had slowed significantly. This left many plots of land available for development until the population began to grow again in the second half of the 19$^{\text{th}}$ century. Thus, between 1668 and 1855, there was an intermittent period when supply was arguably not constrained.

We exploit this time variation to study the extent to which the price effects that we document are driven by periods of constrained housing supply. In Table IV, columns (1) and (3) replicate the specifications of the corresponding columns in Table II. In columns (2) and (4), we add interaction effects

(*Supply-Constrained*) for periods with a highly constrained housing supply (pre-1668 and 1855 to 1884). Note that it might be harder to estimate the effect of lagged birth cohorts in these periods, since past birth cohorts were responsible for a smaller share of aggregate demand due to migration. Importantly, however, we find no evidence that adding these interaction effects changes our main results. In summary, the effect of past birth rates on current house prices is not driven by these periods with significant migration and supply constraints.

In conclusion, while new construction was possible and might have mitigated the price effect of demographic demand shocks, it did not prevent price differences between owner-occupied and rental housing. This suggests that even if an increase in demand for owner-occupied housing was predictable, investors failed to offer enough properties for sale, either previously rented or newly built, by the time this demand materialized.

D. General Asset Demand

An alternative explanation is that the large impact of lagged birth rates is not driven by age-dependent demand for homeownership per se, but rather by time-varying demand for assets more broadly. For example, a substantial body of theoretical literature suggests that young individuals might invest more in risky assets, leading to lower yields on such assets.

Amsterdam presents a useful setting to test for such effects because it already had well-developed capital markets in the premodern period. This implies that early on, households had access to a wide range of investment opportunities compared to other economies. Among all estates registered between the late 17$^{\text{th}}$ and the early 19$^{\text{th}}$ centuries, about 30% of total wealth was invested in real estate, 51% in bonds, 11% in equities, and another 8% in various credit instruments such as mortgages, mortgage-backed securities, and repo loans (Korevaar, 2023). A significant portion of this money was invested internationally.

To test whether lagged birth rates also affect the yields on other assets, Table V replicates the main specifications from Table II for bond and dividend yields. In Internet Appendix Figure IA.7, we plot the coefficients when changing the lags, similar to Figure 4. Bond yields are based on national debt (post-1810) or provincial debt (pre-1810), and dividend yields are for the national Dutch East India Company alone. However, most economic activity was concentrated in Amsterdam (see Section III of the Internet Appendix), and birth rates correlated strongly across areas, implying that the results are unlikely to vary significantly due to geography. Except for the turbulent period around 1800, Dutch (or Holland) debt was perceived as very safe with low rates of return compared to other assets, although somewhat less liquid in the early part of the sample (Gelderblom and Jonker, 2011). Dividend yield data end in 1795 when the Dutch East India Company went bankrupt. Due to the shorter sample, we do not incorporate lagged controls for changes in dividend yields, as adding 14 extra coefficients would lead to overfitting.

Table V

The Effect of Lagged Birth Rates on Bond and Dividend Yields in Amsterdam

This table provides estimation results from equation (1) for bond (i) and dividend yields (d) on lagged birth rates (B) in Amsterdam. We report Newey-West standard errors using a lag length of five. $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Dependent Variable:	$i_t - i_{t-5}$			$d_t - d_{t-5}$	
	(1)	(2)	(3)	(4)	(5)
$B_{t-25/29}$	1.510	1.966	1.555	−0.807	−1.192
	(1.677)	(1.546)	(1.318)	(0.748)	(0.792)
$B_{t-60/64}$	1.860	1.985	1.755	1.255**	1.525**
	(1.187)	(1.340)	(1.413)	(0.634)	(0.751)
Demographic controls	No	Yes	Yes	No	Yes
Economic controls	No	Yes	Yes	No	Yes
Lagged controls (both)	No	No	Yes	No	No
Observations	256	256	256	145	145
Adjusted R^2	0.025	0.048	0.213	0.050	0.076

Looking at the coefficients for birth rates 25 to 29 years ago, we find no significant effects for bond (i) and dividend (d) yields, regardless of whether demographic and economic control variables are included (columns (2) and (4)) or their lags for bond yields (column (3)). We find some evidence that birth rates 60 to 64 years ago have a positive and significant effect on yields for other assets. While the economic magnitudes are similar for both assets, standard errors for bond yields are large, implying that the effect is statistically significant only for dividend yields. This holds both including and excluding control variables. Comparing these effects to rent-price ratios in Table II, we find that the effects on rent-price ratios in all cases are economically larger and in most cases have larger p-values. This holds uniformly when looking at specifications with the most extensive set of controls (columns (3) and (5)), as this results in much more precisely estimated coefficients with standard errors ranging from around 0.5% to 0.7%.

While none of the effects on bond and dividend yields are precisely estimated zeros, the significant difference with the effect on rent-price ratios confirms our intuition that the large effect of lagged birth rates on rent-price ratios is mostly specific to the housing market and driven by age-dependence in the demand for homeownership. Only for birth rate lags corresponding to older ages is there some evidence for a smaller, albeit possibly significant, effect on yields on other assets. This is not surprising, as the effect of lagged birth rates on rent-price ratios for these age groups is likely not only driven by transitions from owner-occupied housing to rental housing but also by individuals passing away and heirs selling their assets. Because equities and (particularly) government bonds are also significant components of older citizens' portfolios, such sales could also apply upward pressure on the yields of these assets.

V. Conclusion

The main conclusion of this paper is that changes in demographics have been a major, predictable driver of variation in house prices over the 17th to 19th centuries, and likely continue to be so today. Importantly, this effect does not appear to be driven by age-dependent changes in the demand for housing consumption, but by a strong age concentration of entry into and exit from homeownership.

Young adults start moving from rentals to homeownership when they settle down in their late 20s and 30s, and become net sellers only when they are older or pass away. When other market participants respond sluggishly to such changes, these demographic changes can lead to large changes in house prices while barely affecting rents. We find various pieces of evidence supporting this channel. First, sale probabilities are slower to adjust to changes in demographic demand to homeownership, suggesting that other market participants start to respond only after prices change. Second, growing demographic demand for rentals pushes up prices more in areas with many investment properties, while growing demographic demand for homeownership increases prices more in areas with a high rate of owner-occupancy. Both effects point to frictions in converting properties from owner-occupied to rental and vice versa, which likely contributes to the price effects we find. Although general age-dependence in asset demand might play a role in explaining the differential effect of demographic shocks on house prices and rents, there is less evidence to support this channel.

In sum, these findings imply that in countries with a rapidly aging population, reduced investment demand for housing will likely result in lower house price growth, when baby boomers retire and gradually pass away. At the same time, many areas are still experiencing an influx of young people, due to both migration and large past birth cohorts. When these cohorts turn to homeownership, that will likely push up house prices.

Initial submission: October 1, 2023; Accepted: August 26, 2024
Editors: Antoinette Schoar, Urban Jermann, Leonid Kogan, Jonathan Lewellen, and Thomas Philippon

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.